Introduction

Goal of this project

- Welcome to Opportunity Plover! This project aims to improve LLM performance in **MCP tool-calling tasks** by using **Reinforcement Learning with Verifiable Rewards** (RLVR). It introduces a **rubric-based reward system** that provides detailed, multidimensional feedback for complex, multi-step reasoning.
- In this project, you will write a prompt that requires the use of a tool(s) to be fulfilled. You will then observe the trajectory the model uses to generate its response. Your goal is to rewrite the prompt until the model generates an incorrect response. Upon model failure, you will create a rubric that not only defines what an ideal response must contain but also the ideal trajectory the model must use to achieve that response. Your work will enhance the ability of cutting-edge LLMs to provide fitting and sophisticated answers to a diverse set of user prompts

We recommend that you review the instructions as you read each section of the course to better understand the specific requirements of each part of the task

Copying and pasting or **using an LLM** to answer questions is not allowed; if you do, you will be removed from the project.

What this module will cover

- Definitions to understand the project
- Task Workflow
- Prompt Requirements + Common Errors and Best Practices
- SOTA Trajectory + Common Errors
- Rubric Requirements + Common Errors and Best Practices
- Video to prepare you for the screening

What Happens After Onboarding?

After completing the onboarding course, your next step is to attend a **War Room** - Opportunity Plover touchpoint after Onboarding Course completion, open Monday through Friday. Look for the invitation in your dashboard!

War Room 🤝

This is your opportunity for **direct interaction with the host and team**. The main purpose is for you to **ask all your remaining questions** so everything is perfectly clear before you take the assessment.

- **Tip:** Please take **notes** of anything that was unclear—whether it's from the course materials or the final recording—and bring those questions with you!
- You can join the War Room at a time that works best for you.

The Assessment & Next Steps ✅

Once you've attended the War Room, the **assessment will be added to your dashboard**.

- **If you pass**: Fantastic! You'll be able to continue working on the project.
- **If you don't pass:** Don't worry! You will be reallocated to another project.

MCP

👋 **Welcome!**
In this project, we are going to train models using the **MCP protocol.**
But wait... do you know what MCP really means? 🤔

👉 **MCP = Model Context Protocol**

- It's a framework that allows AI to understand, connect, and use **external tools**.
- Think of it as a bridge 🌉: AI ↔️ Apps, Data, and Knowledge.
- With MCP, AI becomes not just smart, but also useful in real-world tasks.

💡 In simple words:
MCP helps AI use the right tools at the right time, so it can provide a faster, better, and more accurate answers. ⚡

⚡ How does MCP work?

The MCP Protocol can be explained in 3 simple steps:

Connect → AI links to apps, tools, or data sources. 📂
Understand → AI interprets the information in a clear format. 🧠
Use → AI applies that info to answer, solve tasks, or create results. 🚀

👉 In short: MCP = AI + Tools + Data → Smarter Actions ✅

Back to Outlier

Opportunity Plover

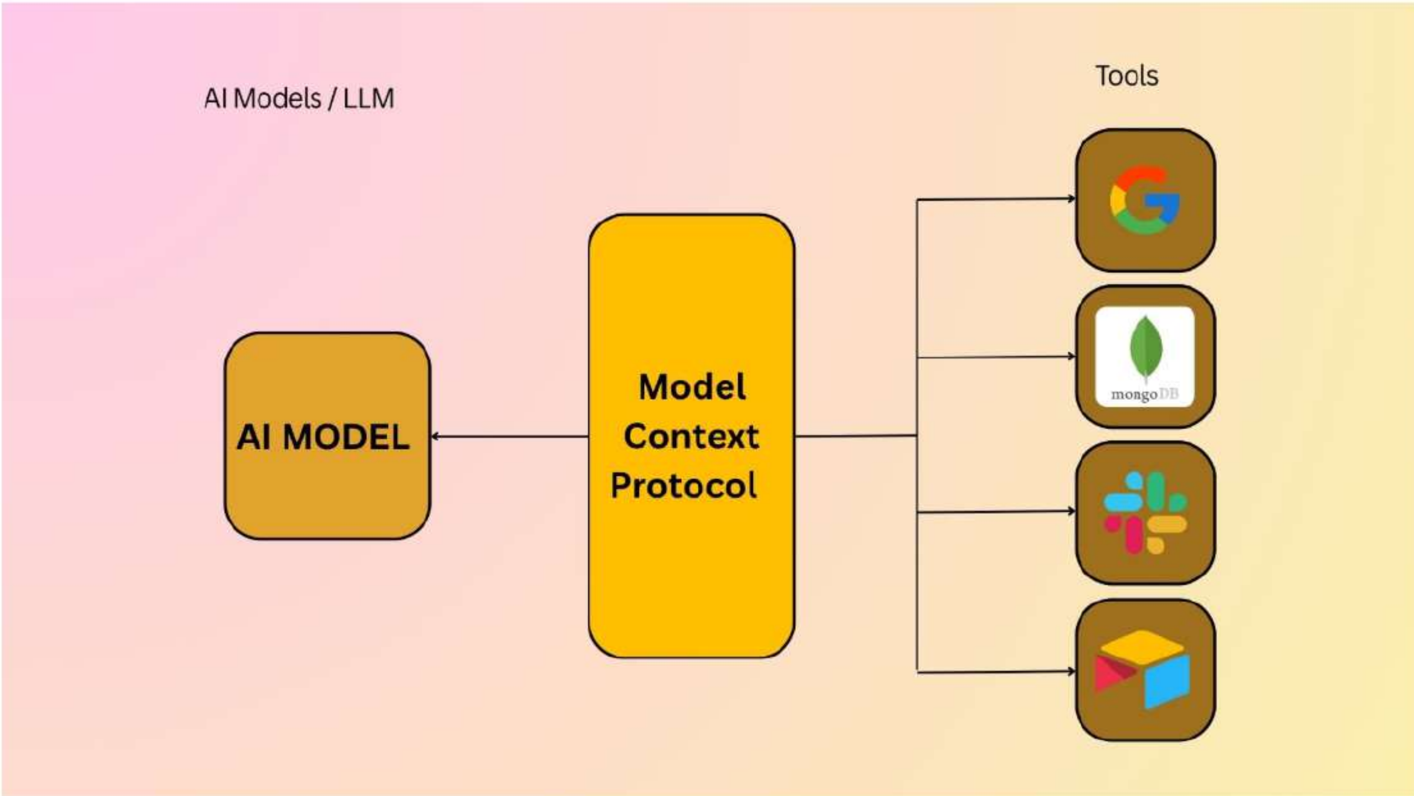
Part 1: Onboarding

Opportunity Plover Onboarding Course
15 sections • 70 min

Opportunity Plover Post-Workshop Quiz
3 sections • 21 min

Opportunity Plover Common Errors
18 sections • 1 min

Part 2: Tasking



Back to Outlier

Opportunity Plover

Part 1: Onboarding

Opportunity Plover Onboarding Course
15 sections • 70 min

Opportunity Plover Post-Workshop Quiz
3 sections • 21 min

Opportunity Plover Common Errors
18 sections • 1 min

Part 2: Tasking

Some definitions before we start

Servers
A server = a place where AI can access different tools (like apps or data sources). 💻

Tools
Tools are external resources that AI can use to extend its abilities.
Think of them like apps or plugins you install on your phone 📱:

- They add new skills to the AI.
- They allow the model to do things it couldn't do alone.
  - 🔹 Examples of Tools:
    - 📂 Access a database
    - 📆 Grab information from a calendar
    - 📩 Send or read an email

What is a tool call?

A tool call is when the model asks a **tool to run by sending a request in JSON format**. It includes two main parts:

- tool → the name of the tool 🛠️
- arguments → the input parameters the tool needs 📄

Example:

```json
{
  "tool": "translate",
  "arguments": {
    "text": "Hello world",
    "language": "es"
  }
}
```

Part 1: Onboarding

Opportunity Plover Onboarding Course
15 sections • 70 min

Opportunity Plover Post-Workshop Quiz
3 sections • 21 min

Opportunity Plover Common Errors
18 sections • 1 min

Part 2: Tasking

Combining Multiple Tools

First Tool Call:

```
{
  "tool": "web_search",
  "arguments": { "query": "Height of Eiffel Tower and Leaning Tower of Pisa (in meters)"
}
```

First Tool Output:

```
{
  "eiffel_m": 324,
  "pisa_m": 56
}
```

Second Tool Call:

```
{
  "tool": "calculator",
  "arguments": { "operation": "subtract", "a": 324, "b": 56 }
}
```

Second Tool Output:

```
{
  "difference_m": 268
}
```

📝 Rubrics in MCP Training

👉 **What are Rubrics?**
Rubrics are evaluation tools that define clear standards for measuring AI responses.

👉 **Why do we use them?**

- ✅ To evaluate the model's performance step by step.
- ⚖️ To ensure consistency across different evaluations.
- 📊 To score answers based on different criteria.
- 🚀 To help models improve through structured feedback.

💡 In short:
Rubrics = a scorecard that makes AI training fair, consistent, and effective.

Interactive Data Explorer

Think of this app as your **map and magnifying glass for our new toolset**.

What Is It For?

Your goal is to write prompts that test the model's ability to use tools and then, when it fails, to create a perfect "golden rubric" for the ideal response and tool-use trajectory. This explorer is essential for that job.

You will use it to:

1. **Discover:** See exactly what data and tool functions are available.
2. **Plan:** Find the specific data you'll need to write a challenging prompt.
3. **Create Rubrics:** When a model fails, you'll use this tool to find the ground truth—the exact data chunk or API response the model should have found. This "ground truth" is what you will use to define the ideal trajectory in your rubric.

Interactive Data Explorer

Interactive Data Explorer

Interactive Data Explorer

Think of this app as your **map and magnifying glass for our new toolset**.

What Is It For?

Your goal is to write prompts that test the model's ability to use tools and then, when it fails, to create a perfect "golden rubric" for the ideal response and tool-use trajectory. This explorer is essential for that job.

You will use it to:

1. **Discover:** See exactly what data and tool functions are available.
2. **Plan:** Find the specific data you'll need to write a challenging prompt.
3. **Create Rubrics:** When a model fails, you'll use this tool to find the ground truth—the exact data chunk or API response the model should have found. This "ground truth" is what you will use to define the ideal trajectory in your rubric.

Interactive Data Explorer

⚙ How to Use the Explorer

Method 1: The Search Function (for finding specific keywords)

This is your "quick find" tool. Use it when you have a specific piece of text, value, or key in mind.

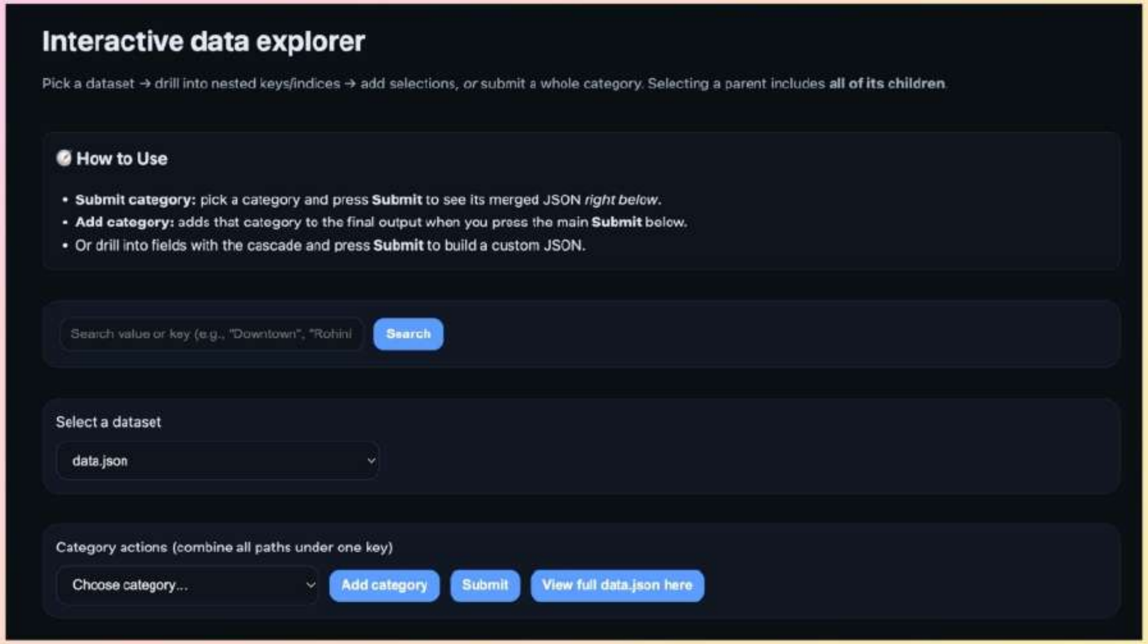
- **When to use:** You're looking for a specific item, like "Downtown," "Rohini," or "Gym."
- **How it works:**
    a. Type your keyword into the **"Search value or key"** box.
    b. Click the **Search** button.
    c. **Result:** The app will instantly filter the entire dataset and show you only the JSON chunks that contain your keyword.
        i. **If multiple matches exist** → you'll see only those chunks.
        ii. **If only one match exists** → the full JSON file is shown.

Method 2: Selecting & Exploring Datasets (For broad exploration)

This is your "deep dive" tool. **Use it when you want to understand the structure of a dataset or combine different parts to build a custom output.**

- **When to use:** You need to see all "apartments" or combine "email" and "contacts" into one file.
- **How it works:**
  a. **Select a dataset:** Choose the main JSON file from the **"Select a dataset"** dropdown.
  b. **Explore Categories:** Now, look at the **"Category actions"** section. This is where you can drill down.
  c. You have two options here:
     - **Option A:** View **one** category
       - Choose a category from the dropdown (e.g., "apartments").
       - Click the **Submit** button.
       - **Result:** You'll see the merged JSON for only the "apartments" category.
     - **Option B:** Combine **multiple** categories
       - Choose a category (e.g., "apartments").
       - Click the **Add category** button.
       - Choose another category (e.g., "city").
       - Click **Add category** again.
       - When you've added all the categories you want, click the **Submit** button.
       - **Result:** You'll get a single, combined JSON file with both "apartments" and "city" in it.
  d. **Copy / Download:** Export your final JSON.

This is how the interactive data explorer will look like, you will find the URL to access it in the task



Task Requirements

Before you start....

When you open the task, the first thing you'll see are four key items (requirements):

- **Task Complexity:** 10+ Tools
- **Number of Unique Servers:** 2
- **Servers to Target:** n/a
- **Domain:** Work

- **Task Complexity** ⚒️: minimum number of tools required to complete the task.
- **Number of Unique Servers** 🗄️: number of servers you should use to select the tools.
- **Servers to Target** 🎯: servers that must be used at least once in this task.
- **Domain** 🌐: the topic or field the prompt should be about.

Overview of a Task

Task Steps

A task on this project consists of these steps:

1. **Set Up The Tool Environment**
   - Before writing your prompt, you should get thoroughly familiar with the Server the task requires you to use. This is where the Interactive Data Explorer and the task Playground come in handy.
2. **Write a prompt**
   - When writing a prompt, you must ensure the following attributes:
     - Tool-dependent 🛠️
     - Timeless 🕰️
     - Unambiguous 🎯
     - Conversational 💬
     - Subtle in tool request
     - Feasible
     - Cause an SOTA (state-of-the-art) failure

- Feasible
- Cause an SOTA (state-of-the-art) failure
- Have a single GTFA (Ground Truth Final Answer)

3. **Check for model failure**
   - Your job is to evaluate the model's reasoning, tool use, and final response to flag if the model failed (does not align with the GTFA - Ground truth final answer) and what type of failures are present.
4. **Record your "Silver Trajectory" in the Agent Execution Playground**
   - Once you have a prompt that successfully generates a SOTA model failure, you will need to go back to the Agent Execution playground to add all the tool calls required to fully answer your prompt and get rid of any unnecessary tool calls. The final result should be a series of tool calls that lead to your GTFA, i.e., the "silver trajectory". In step 6, the trajectory you record here and your provided GTFA will be graded against your rubric criteria.
5. **Create A Rubric**
   - A good rubric is a simple and reliable checklist to evaluate how well the model performed a task. Rubrics provide binary criteria that map directly to what the model was expected to do, based on the prompt, final answer, and tool outputs. A high-quality rubric is objective, grounded in the task, and easy to apply consistently.
6. **Review LLM-generated scores against your rubric**
   - In this step, you will review LLM-generated scores for:
     - a) Your playground trajectory and GTFA
     - b) The SOTA trajectory and final response against your rubric
   - If there are any ratings you disagree with, you will have to make a note of them in the provided fields.
7. **Select Task difficulty**

Part 1: Onboarding

Opportunity Plover Onboarding Course
15 sections • 70 min

Opportunity Plover Post-Workshop Quiz
3 sections • 21 min

Opportunity Plover Common Errors
18 sections • 1 min

Part 2: Tasking

- Once you have a prompt that successfully generates a SOTA model failure, you will need to go back to the Agent Execution playground to add all the tool calls required to fully answer your prompt and get rid of any unnecessary tool calls. The final result should be a series of tool calls that lead to your GTFA, i.e., the "silver trajectory". In step 6, the trajectory you record here and your provided GTFA will be graded against your rubric criteria.

5. **Create A Rubric**

   - A good rubric is a simple and reliable checklist to evaluate how well the model performed a task. Rubrics provide binary criteria that map directly to what the model was expected to do, based on the prompt, final answer, and tool outputs. A high-quality rubric is objective, grounded in the task, and easy to apply consistently.

6. **Review LLM-generated scores against your rubric**

   - In this step, you will review LLM-generated scores for:
     - a) Your playground trajectory and GTFA
     - b) The SOTA trajectory and final response against your rubric
   - If there are any ratings you disagree with, you will have to make a note of them in the provided fields.

7. **Select Task difficulty**

   - You will need to select how long it would have taken you to answer the prompt without the model's help

We will cover the main steps in more detail throughout the rest of the course.

Back to Outlier

Opportunity Plover

Part 1: Onboarding

Opportunity Plover Onboarding Course
15 sections • 70 min

Opportunity Plover Post-Workshop Quiz
3 sections • 21 min

Opportunity Plover Common Errors
18 sections • 1 min

Part 2: Tasking

Part 1: Prompt Creation

Part 1: Prompt Creation

Select the tools you want to use 🛠️

- Pick the main tools that will solve your task.

Add distraction tools 🎭

- Include some tools that you don't really need, just to test if the model stays focused.

Test in the Playground 🎮

- Run each tool and check their outputs.
- Understand what kind of results you get.

Note: Make sure to review the **Set Up The Tool Environment** section in the instructions to learn how to use the playground and the specific requirements related to the tools

Build the Prompt 📝

- Write a prompt that forces the model to use the selected tools.
- **Make sure the answer can only be correct if the model calls those tools.**

Part 1: Onboarding

Opportunity Plover Onboarding Course
15 sections • 70 min

Opportunity Plover Post-Workshop Quiz
3 sections • 21 min

Opportunity Plover Common Errors
18 sections • 1 min

Part 2: Tasking

The answer to a prompt should not be time-sensitive. In some cases, the answer to the prompt may gradually evolve over time, but that doesn't necessarily make them invalid. The goal is to avoid information that changes frequently (e.g., daily or weekly), not to reject any prompt with potential future variation.

Prompt Dimension to Check	Prompt Examples
Timelessness	✅ **Acceptable examples - DO NOT COPY THESE:** - "Back in July 2023, I had $5,000 in savings. A friend suggested I invest in Apple stocks, as it was performing well at the time. Instead, I invested in NVIDIA stocks. I remember buying them on July 5th and selling them a day before Christmas the next year at the closing price. I'm wondering if I would have earned more had I followed my friend's advice. Can you tell me how much money I would have gained or lost if I took the suggestion?" —> **The prompt is focused on a specific historical scenario and does not rely on current or future events, making it applicable regardless of when it is analyzed.** - "I recall when Exblifep got approved for treating complicated urinary tract infections by the FDA, and I was convinced the market was going to boom. Out of excitement, I decided to buy 100 shares of Anuh Pharma from the Bombay Stock Exchange on the same day Exblifep got its approval for treating complicated urinary tract infections by the FDA, since I only trade on the BSE. But just as I was about to place the order, my friend stepped in and strongly advised me to go for Syncom Formulations instead. I got confused by his advice and ended up buying Syncom Formulations shares at their closing price. Later, when I sold them on June 27, 2024, at their closing price, I faced a huge loss. Now I can't stop wondering if I had stuck to my original plan, bought 100 shares of Anuh Pharma at their closing price, and sold them on June 27, 2024, at their closing price, my loss might have been smaller, and I wonder by how much." —> **The prompt is structured in a way that does not rely on current events, making it applicable regardless of the time period in which it is used. It focuses on a hypothetical scenario and personal reflection, which are not bound by time-sensitive details.** × **Invalid example:** Asking for the number of comments on a post is not valid, as they change over time.

A valid prompt for this project **must require tool calls to be answerable**. It must be a prompt that cannot be answered with general knowledge alone that any LLM may have.

💡**Pro tip:** If the SOTA model answered the prompt correctly without using a tool call, then your prompt is incorrect, as it does not require a tool, even if you planned on using one.

Prompt Dimension to Check	**Prompt Examples**
Tool-Dependent	✅ **Acceptable examples - DO NOT COPY THESE** : - "I'm planning a trip to Mexico City and I need a list of 3 5-star hotels that have a swimming pool among their amenities and bathrooms with jacuzzis." - "I work at the car dealership, and my boss asked me to use the database to calculate the mean total unique visitors per year for the driving test page of our website between 2020 and 2023. Can you help me with this?" - "I was reading the PythonOneLiners repository of Finxter, and I found it amazing. My favorite code example was in the Python trick 9 file, but I am having trouble executing it. Can you execute it for me and give me the output?" —> **The prompt explicitly requests the execution of a specific piece of code, which requires a tool or environment capable of running Python code to fulfill the request.** - "Look into the database for the cheapest ground floor apartment with premium amenities, deposit less than 1500, and built in or after 2022. Create a simple text file using the apartment as the file name containing the rate and who's managing it as of August 2025." —> **This prompt is evidently tool-dependent because it requires specific actions that are reliant on the capabilities of a particular tool or system, such as accessing a database and creating a text file.** × **Invalid example**: - "Can you help me with a list of all the Academy Awards that Titanic won in 1997?" —> **No tool use required because LLMs are trained with this data.**

Part 1: Onboarding

Opportunity Plover Onboarding Course
15 sections • 70 min

Opportunity Plover Post-Workshop Quiz
3 sections • 21 min

Opportunity Plover Common Errors
18 sections • 1 min

Part 2: Tasking

A valid prompt for this project **must have only one clear and correct answer: the GTFA**. The GTFA should be based on the tools' outputs. Your prompt fails if it has multiple valid final answers, so avoid asking open-ended questions or giving the model the reins to make assumptions.

Prompt Dimension to Check	Prompt Examples
Unambiguity **Avoid vague requirements such as:** × *Give me an overview* × *Show me some examples* × *Briefly describe* × *Find me some options* × *Provide a summary*	✅ **Acceptable example - DO NOT COPY THESE :** - "I'm conducting a literature review about vaccine development. Please find all the research articles authored by "Katalin Karikó" in 2021 related to "mRNA vaccine technology," and provide its complete title, journal name, and publication date." - "My friend QT posted a movie recommendation in the group chat not too long ago. He told me to watch it if I haven't, so I will have to check in the movie list I've created to keep track of films I've seen. I plan to watch it tomorrow between my appointments, but I only have a couple of hours in between them. Can you check my list and see whether I will be able to watch this film in between appointments, or should I just watch it after my appointments?" —> **The prompt clearly outlines the task of checking a movie list to determine if there is enough time to watch a recommended film between appointments, leaving little room for misinterpretation.** - "I keep a register of my crypto transactions in my local data records, and I'm analyzing new coins other than Bitcoin and Ethereum. On the date when I received stellar native crypto according to my transaction records, I was actually deciding between that and Dogecoin. If instead I had chosen Dogecoin, what would the closing price difference have been between Dogecoin and Stellar exactly one year later?" —> **The prompt clearly specifies the information needed, including the context of the transaction records, the specific cryptocurrencies involved, and the exact time frame for the price comparison, leaving little room for misinterpretation.** - "Can you help me list all CSV files on my computer that have missing data so I can test out my missing data imputation located in my public code, and avoid generating any code or script for this?" —> **The prompt clearly specifies the task of listing CSV files with missing data without generating any code or script, leaving little room for misinterpretation.** × **Invalid example:** - "I'm conducting a literature review about vaccine development. Please find the single largest research article published by "Katalin Karikó" in 2021 related to "mRNA vaccine technology," and provide its full PubMed ID, complete title, journal name, and publication date." —> **What does it mean by "Largest"? Word count? Page count? Citations?** - "I work at a car dealership, and I have been tasked with analyzing the employee performance from 2015 to 2023. I need to identify the top 3 performing departments based on their average total scores across all years. Additionally, highlight the year with the highest overall employee performance and explain what metric contributed the most to that peak." —> **Doesn't define what metrics are being measured (sales numbers, customer satisfaction, service times, etc.). Should the metric be the one that increased the most relative to other years, or the one with the highest absolute value?**

Part 1: Onboarding

Opportunity Plover Onboarding Course
15 sections • 70 min

Opportunity Plover Post-Workshop Quiz
3 sections • 21 min

Opportunity Plover Common Errors
18 sections • 1 min

Part 2: Tasking

A valid prompt for this project **must not mention the tools or the servers needed to answer it.** The prompt must sound like something a real user would ask. A real user would not know the names of tools or the underlying services and URLs being used by the model.

Thus, **the prompt must not mention tool names explicitly** (e.g., "get_weather", "search google"), server names (e.g., "Airtable", "MongoDB"), internal data sources (e.g., "Filesystem"), or specify URLS(e.g., "apple.com/etc", "google.com").

Prompt Dimension to Check	Prompt Examples
Tools are not directly mentioned	✅ **Acceptable examples - DO NOT COPY THESE :** - " I have a file in my data folder that contains all my old trades. I want to look back at my very first Bitcoin purchase to see if I got a good deal. Could you locate the first BUY transaction for BTC in the file, and then use a financial data service to determine the official opening price for BTC/USD on that same day? I want to see how the spot price I paid compares to the market's opening price." —> **The prompt subtly requests the use of a financial data service to find the official opening price for BTC/USD, without explicitly stating which tool or service to use, allowing for flexibility in how the task is accomplished.** - "What is the difference in the total purchase amount between members and non-members before August 1, 2025, in the customer table of the video game store, according to the database?" —> **The prompt implicitly requests a database query or analysis without explicitly stating the need for a specific tool or method, focusing instead on the information required.** - " I am working on a campaign, and I need to know the average amount paid by customers whose IDs had four zeros in them. All I remember is that the store was related to video games. Can you help me by searching for it?" —**> **The prompt indirectly asks for a specific action (searching for information) without explicitly stating the use of a tool or method, making the request less direct.** × **Invalid example:** - " I want to focus my marketing strategy on increasing the average ticket per sale for my video game store. I have a record of purchase history segmented by customer regularity in MongoDB, and I would like to start by increasing the amount from the lowest average ticket. Which customer segment should I target first according to my records? " —> **Prompt mentions MongoDB, which is the tool server.**

A valid prompt for this project **must sound like something a real user would ask**. Use **natural**, **conversational language** and provide **a believable reason** for the request.

Prompt Dimension to Check	Prompt Examples
Conversational	✅ **Acceptable examples - DO NOT COPY THESE :** - "I was trying out a new trading strategy last summer and invested in a couple of car companies. I bought 100 shares of Tesla on June 2, and then 20 shares of Ford on June 10, 2025. I decided to close out both positions on July 30th. Can you figure out what my total profit or loss was from those two trades?" —> **The prompt is written like a little story — it shares a personal experience and asks for help in a friendly, engaging way. That's what makes it feel conversational.** - "At the end of July 2025, I was catching up with a colleague of mine at La Mar, and he explained that he had been very successful at day trading by buying stocks at market opening and selling them at close, choosing stocks of foreign companies that trade on the NYSE. His logic was that, since the companies were far away, any bad news that could lower the stock price would only be reflected the following day. He also told me that, on that very same day, he had managed to make over 2% on a Colombian bank called Bancolombia. Now I want to believe him, but he tends to exaggerate a bit, so was he telling the truth?" —> **The prompt is written like a story, almost like a natural conversation where someone shares an anecdote and asks for validation or advice. That makes it feel more engaging and relatable.** × **Invalid example:** - "Please provide the following information regarding Apple Inc.'s acquisition of a gaming development studio in May 2025: The name of the acquired gaming studio and the date when the acquisition was first publicly reported. The impact of the acquisition announcement on Apple Inc. (AAPL) stock price, more specifically: Opening share price on the acquisition announcement date, closing share price on the acquisition announcement date, percentage change between opening and closing prices." - "On June 2, 2025, I purchased 100 shares of Tesla. On June 10, 2025, I purchased 20 shares of Ford. Both positions were closed on July 30, 2025. Calculate the total profit or loss resulting from these transactions." - "A movie recommendation was posted by QT. Determine whether the recommended movie can be viewed within the available time slot of two hours between existing appointments, based on the user's preexisting list of watched films. If not, indicate that the movie should be scheduled after the appointments."

Write your first prompt based on the **Slack** server. Ensure your prompt is complex enough to use both tool calls. It should also be complex enough to **challenge SOTA**. We'll also give you some channel information.

Here's a detail of the server:

channel_id	channel_name
C0938PV4E7Q	#gaming-suggestions

UserID	UserName	RealName	Channel	Text
U093YBBLH9U	hiphopluvr1989	Omari West	C0938PV4E7Q	Def Jam Fight For New York was a classic growing up
U094G2BTX7B	shinsplints7070	Steve Shins	C0938PV4E7Q	Apex Legends is my favorite battle royale game
U0944N06J20	quickfits01	James Fitz	C0938PV4E7Q	I love sports games too Im always buying the new FIFA

Server Name	Tool Name	Parameters	Output
slack	channels_list	channel_types, cursor, limit, sort	Returns a CSV with all the channels in the server. channel_type can be public_channel or private_channel
slack	conversations_history	channel_id, cursor, include_activity_messages (checkbox), limit	Returns a CSV with channel messages

Prompt: Common Errors and Best Practices

These are the most common errors we've found based on quality reads when reviewing prompts. We've included a brief explanation of each error and the best way to avoid them or improve the prompt.

Prompt: Ambiguity (Multiple Answers)

The prompt uses vague terms ("best," "summarize"), allowing for multiple valid interpretations.

✅ Best Practice: **Be Precise and Specific**
Your prompt must be a question with **only one possible correct answer**. Define all subjective terms with objective metrics.

- Treat the AI like a brand-new assistant who is brilliant but has zero context. Be explicit!

Example	Prompt
❌ **Vague:**	"Calculate the total purchases from that day." (Fails because it's unclear which data source (collection) to use.)
🎯 **Precise:**	"Using the **'Purchase History' collection**, calculate the total dollar amount of all purchases made on **December 23, 2023**."
❌ **Vague:**	"Find an apartment in a peaceful city with a low cost of living." (Fails because "peaceful" and "low" are subjective.)
🎯 **Precise:**	"Find the apartment with the **lowest violent crime rate** and a **monthly rent under £1500**."

Prompt: Unnatural or Contrived

The prompt reads like a robotic list of instructions, a logic puzzle, or a database query, not a natural, conversational request.

✅ Best Practice: **Write Conversationally with Context**
Write as if you're sending a message to a colleague. Provide a brief backstory to make the request logical and natural.

Example	Prompt
❌ **Contrived:**	"Retrieve file." "Analyze for errors." "List errors."
🗣️ **Natural:**	"**Could you look at 'Q4_Report.csv'?** I need you to find any rows where the 'Total' column is less than the 'Cost' column and list the 'Order ID' for each."

Prompt: Insufficient Complexity

The prompt is too simple and can be answered with a single tool call, failing to test the AI's reasoning.

✅ Best Practice: **Require Multi-Step Reasoning**
Design prompts that force the AI to connect data. Make it find "Fact A" from one tool and "Fact B" from another, then analyze or compare them.

Example	Prompt
❌ **Simple:**	"How many sales did X store had in September, 2018?"
🧠 **Complex:**	"First, find all employees in the **'Sales' department.** Then, find all attendees of the **'Q4 Kickoff' calendar event**. Finally, list any sales employees who were **not on the attendee list**."

Prompt: Not Timeless (Time-Dependent)

The prompt uses relative time references ("today," "last year," "recent") that change depending on when the prompt is run.

✅ Best Practice: **Anchor Your Prompt in Time**

Always use specific, fixed dates and date ranges to ensure the answer remains static based on the available info in the data.

- Ask yourself, "If another person ran this exact prompt five years from now, would they get the exact same answer?" If not, anchor it.

Example	Prompt
❌ **Relative:**	"What was the inflation rate last year?"
⏳ **Anchored:**	"What was the US inflation rate for the **full calendar year 2023?**"
❌ **Relative:**	"Find me recent conversations."
⏳ **Anchored:**	"Find all conversations from **October 2024.**"

Part 2: Evaluate SOTA

Part 2: Evaluate the SOTA model 🚀

SOTA = State of the Art, meaning the most advanced model available for the task.

⚠️ **The SOTA model must fail.**
Later, you'll learn the difference between a valid fail ✅ and an invalid fail ❌.

👉 For now, keep in mind: **if your prompt doesn't make the SOTA fail, you'll need to rewrite the prompt until it does** ✍️.

📌 Once you get the fail, you must briefly **classify the error and describe it** 📝.

What's considered a valid SOTA failure?

To put it simply, a valid SOTA failure is when the **model's response contains an incorrect final answer.**

✅ **Valid** SOTA failures:

- Final response is absent (no response generated or incomplete output)
- Failing to address any of the essential asks (a necessary part of the answer is missing/incorrect)
- Wrong format or structure → only when specific formatting was explicitly required

If the model ultimately produces the correct final answer, then the following do not count as valid SOTA Failures:

- Inefficient trajectories
- Trajectory issues
- Trajectory contains hallucinated data
- Not fulfilling optional/nice-to-have rubrics
- The model response has irrelevant information, factual inaccuracies, or hallucinated information

Please note: If the model asks a follow-up question or the final answer is incorrect because the prompt was ambiguous or vague in any regard, this is not a model failure. If the prompt clearly contains all the information necessary to be fulfilled with the help of the enabled tools, then a model follow-up question is a valid SOTA failure.

Part 1: Onboarding

Opportunity Plover Onboarding Course
15 sections • 70 min

Opportunity Plover Post-Workshop Quiz
3 sections • 21 min

Opportunity Plover Common Errors
18 sections • 1 min

Part 2: Tasking

× Not A Valid Fail Example

Prompt

I'm planning a weather study at the Shenandoah National Park (Virginia). For the study to go smoothly, can you tell me if there is a weather station within 30 km of the national park?

Model's Response

Assistant message:

Assistant

✅ Requirement 1: Weather Station within 30km - SATISFIED

I found several weather stations near Shenandoah National Park:

- **New Market Airport (K8W2)**: 17.2 miles (27.7 km) - **WITHIN 30km**
- Charlottesville-Albemarle Airport (KCHO): 24.8 miles (39.9 km) - also within 30km
- Additional stations at 28+ miles

The closest station at New Market Airport is well within your 30km requirement at just 27.7km from the park.

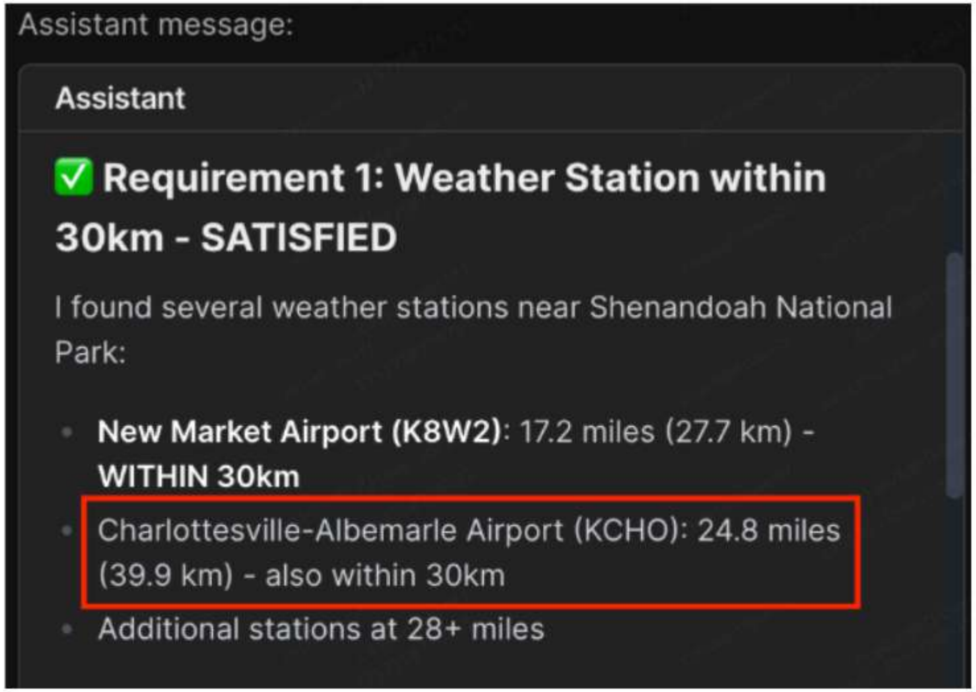
Even though the model wrongly includes the weather station at Charlottesville-Albemarle Airport as a weather station within 30km of the Shenandoah National Park - this **IS NOT A FAILURE.** The prompt asks if there is a weather station within 30 km, and the model correctly states there is one.

✅ A Valid Fail Example

Prompt

I'm planning a weather study at the Shenandoah National Park (Virginia). For the study to go smoothly, can you list all the weather stations that are within 30km of the national park?

Model's Response



Assistant message:

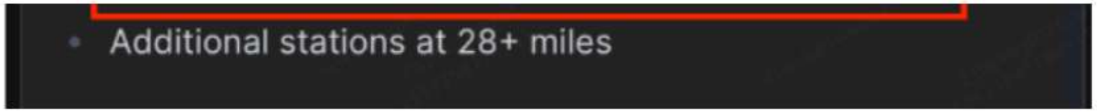
Assistant

✅ Requirement 1: Weather Station within 30km - SATISFIED

I found several weather stations near Shenandoah National Park:

- **New Market Airport (K8W2)**: 17.2 miles (27.7 km) - **WITHIN 30km**
- Charlottesville-Albemarle Airport (KCHO): 24.8 miles (39.9 km) - also within 30km
- Additional stations at 28+ miles

Please observe how important the wording of the prompt is. This prompt is very similar to the first example; however, since the prompt asks to list **all** weather stations within 30 km of the national park, this response is considered **A Valid SOTA Failure.** As you can see, Charlottesville-Albemarle Airport (KCHO) is 39.9 km from the national park, and the model incorrectly includes it in the list. This makes the final answer incorrect - hence a **Valid SOTA Fail.**



Please observe how important the wording of the prompt is. This prompt is very similar to the first example; however, since the prompt asks to list **all** weather stations within 30 km of the national park, this response is considered **A Valid SOTA Failure.** As you can see, Charlottesville-Albemarle Airport (KCHO) is 39.9 km from the national park, and the model incorrectly includes it in the list. This makes the final answer incorrect - hence a **Valid SOTA Fail.**

Key triggers: how to make SOTA fail

Multi-criteria verification challenges

- Combine 2-3 requirements where one is clearly met, but another is borderline
- Models often focus on the easier requirement and make errors on the harder one
- Ask "Does X satisfy BOTH requirements?" - models may affirm prematurely after checking only one
- Including similar distractor tools that seem like they're able to be used to get the right answer, but can't.

Prompt Structure

- Embed the actual requirements late in the prompt after context-setting
- Use technical-sounding studies to make the model less likely to double-check
- Make the model have to strongly reason what tool to use from the enabled tool environment to get the correct final answer.